

Aligning to the Next Generation Science Standards (NGSS)

A new era of science education

Traditional instructional methods in science involve having students memorize disconnected facts or work to confirm established science ideas. But in today's paradigm, teachers are called upon to help students dive deep into scientific exploration, the way real scientists do. This way, students grow as critical thinkers and active learners and are well-prepared to participate in a global society.

The Next Generation Science Standards (NGSS) and other state guidelines represent more than just new expectations: They embody a transformative vision for science education. This shift began over a decade ago, with the 2012 publication of **A Framework for K–12 Science Education** (known simply as the *Framework*), which laid out a bold new vision for science teaching and learning. The vision of three-dimensional teaching and learning became the basis for subsequent standards, such as the NGSS and other state-adopted, *Framework*-based benchmarks.

These benchmarks lay the groundwork for three important shifts away from traditional science instruction, including:

1. Moving students from *learning about* science to *figuring out* science.
2. Teaching students to explain phenomena happening in the natural world.
3. Laying out expectations for three-dimensional learning, as defined below.



Three-dimensional learning

NGSS combine essential domains to make up a three-dimensional approach to science education:

- Science and Engineering Practices (SEPs)
- Crosscutting Concepts (CCCs)
- Disciplinary Core Ideas (DCIs)

These three areas of focus work together to provide students with a well-rounded understanding of science concepts, what it means to think scientifically, and how science is relevant to the world around them. You can learn more about three-dimensional learning with these additional resources:

- [What is three-dimensional learning?](#)
- [The NGSS student experience](#)

The importance of high-quality instructional materials

When you're equipped with an understanding of NGSS, you can make more informed decisions about the curriculum that best suits the needs of your students. **[Choosing high-quality instructional materials \(HQIM\)](#)** means ensuring that the publishers you're considering have:

- Grounded the curriculum in evidence-based standards (such as NGSS or other research-based state standards).
- Conducted unbiased research to determine effectiveness.
- Received independent reviews from trustworthy, independent evaluators (such as EdReports) to confirm that the instructional design meets the intended learning goals.
- Collected customer feedback, such as case studies and testimonials.

When evaluating science curricula, remember that HQIM are always evidence-based. They help educators support student achievement by focusing on the critical skills and knowledge students need to succeed in the science classroom...and in the real world.

To learn more about high-quality instructional materials for science, visit amplify.com/science.

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